CIS 530: ADVANCED DATA MINING

Team Project

Title: "Critical Applications: Predicting Stock Market Trends Through Sentiment Analysis and Machine Learning“

By :
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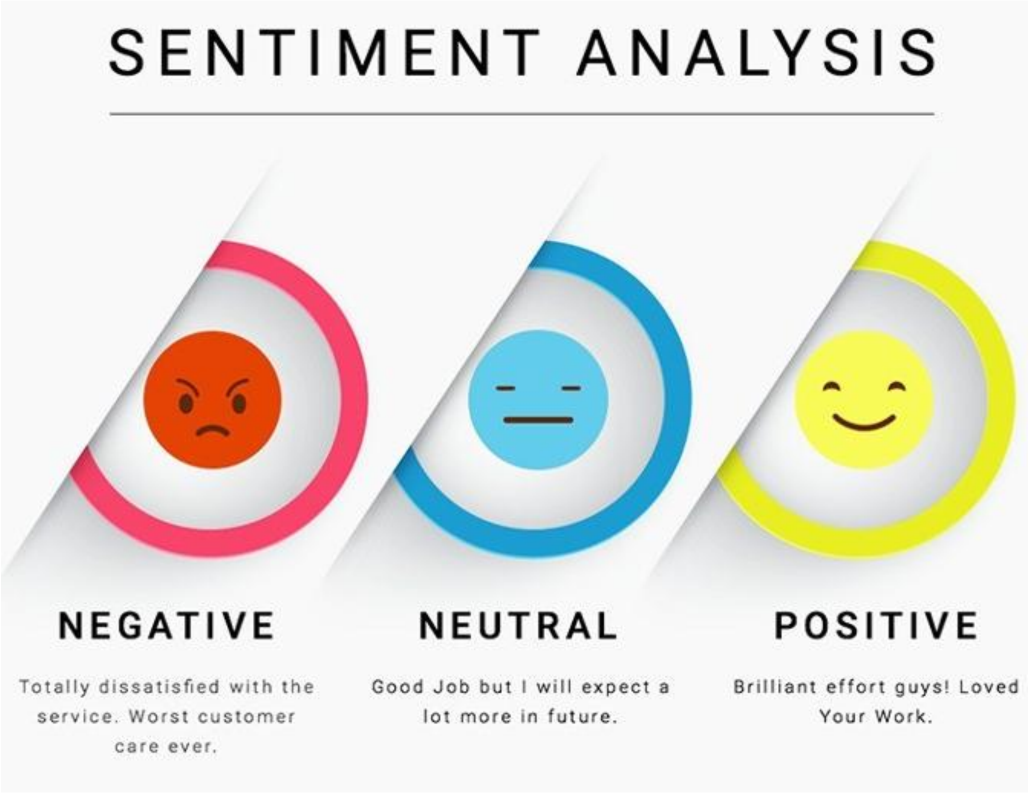
Introduction to Stock Market Prediction through Sentiment Analysis



Objective:

- Predict stock market trends using sentiment analysis of financial news headlines.
- Combine sentiment data with historical stock prices to forecast market movements (up, down, stable).

Key Techniques:

- Sentiment analysis using Vader Sentiment and TextBlob.
- Machine learning models: Logistic Regression, Random Forest, LSTM (Long Short-Term Memory) networks for time-series prediction.

Dataset:

- Financial Sentiment Analysis Dataset (from Kaggle) and Yahoo Finance Data.

The Power of Sentiment in Financial Markets

What is Sentiment Analysis?

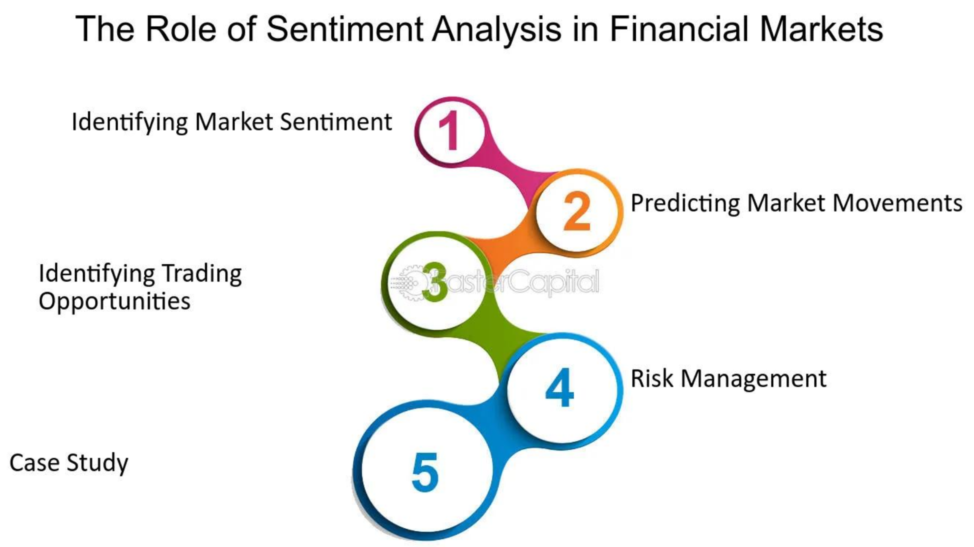
- Sentiment analysis extracts emotional tone from text.
- Financial news can influence stock prices directly as investors react to positive or negative sentiments.

Examples:

- **Positive Sentiment:** “Tech stocks soar after record earnings” → Stock price rises.
- **Negative Sentiment:** “Global recession fears rise” → Stock price falls.

Why It Matters:

- Sentiment provides context beyond hard data, adding a psychological dimension to market predictions.



Related Work: Sentiment Analysis in Stock Market Forecasting

Previous Research:

• **Bollen et al. (2011):** Explored the impact of **Twitter sentiment** on stock market movements.

• **Hu et al. (2020):** Analyzed **financial headlines** using machine learning models like **Support Vector Machines (SVM)**.

Challenges in Prior Research:

• Limited to **traditional machine learning** and lacked advanced deep learning methods like **LSTM**.

• No integration of both **numerical stock data** and **sentiment analysis**.

Problem Definition: Predicting Stock Trends

CORE RESEARCH PROBLEM:

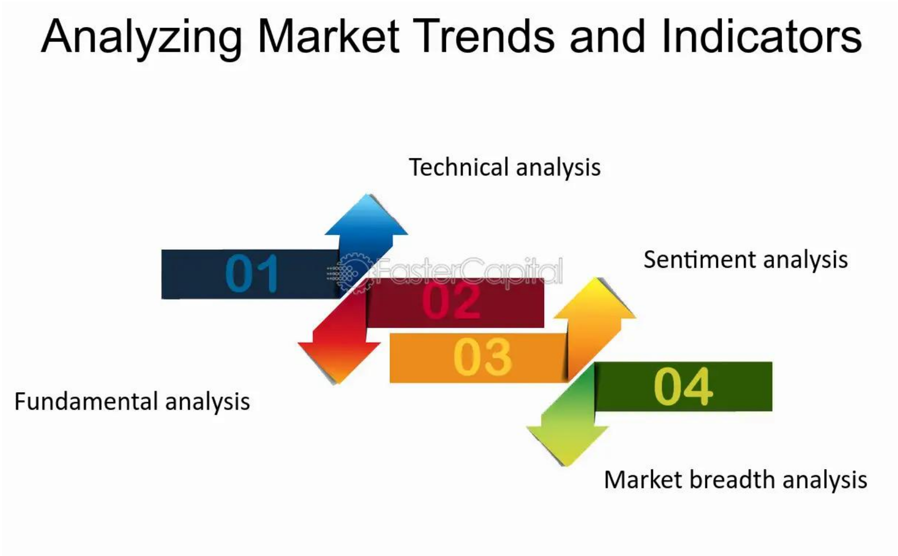
•Can we predict stock market trends (up, down, or stable) using both sentiment and numerical stock data?

KEY CHALLENGES:

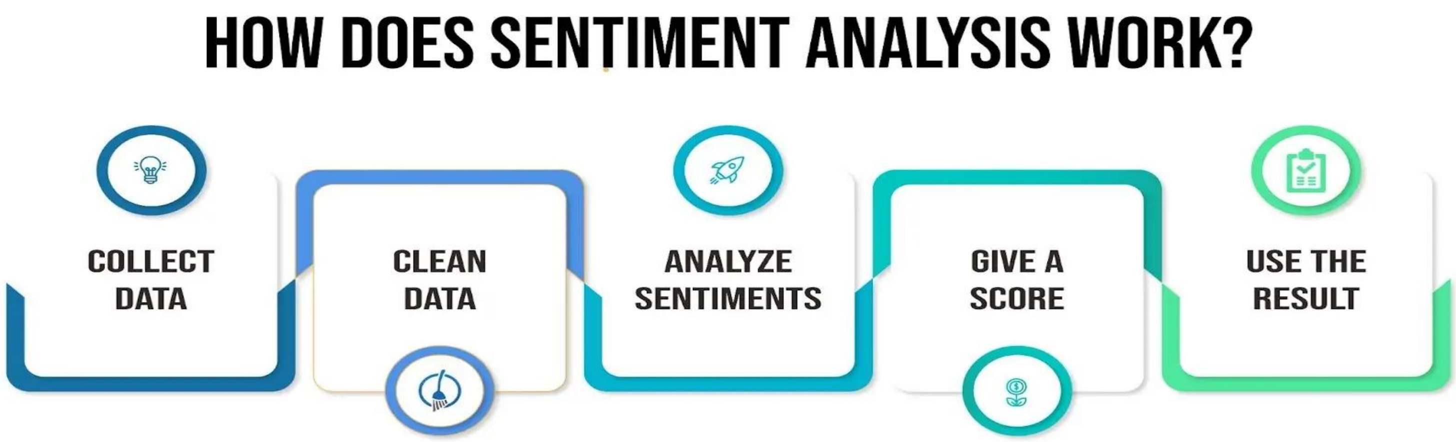
1.**Data Alignment:** Aligning sentiment data with stock price movements over time.
2.**Sentiment Ambiguity:** Headlines with mixed sentiment (positive and negative).
3.**Time-Series Data:** Stock prices are sequential, needing models that can capture these long-term dependencies.

GOAL:

•Build a robust model that combines both **sentiment analysis** and **numerical stock data** to predict market trends.



Methodology - Extracting and Processing Sentiment



SENTIMENT ANALYSIS TOOLS:

- **Vader Sentiment:** Analyzes emotive text and is perfect for headlines.
- **TextBlob:** Provides both polarity (positive/negative) and subjectivity (factual/opinionated).
- **Sentiment Scores:**
- Extract sentiment from financial news and assign a score for positive, neutral, or negative sentiment.
- **Importance of Sentiment Data:**
- Captures emotional tone from the news, directly influencing investor behavior.

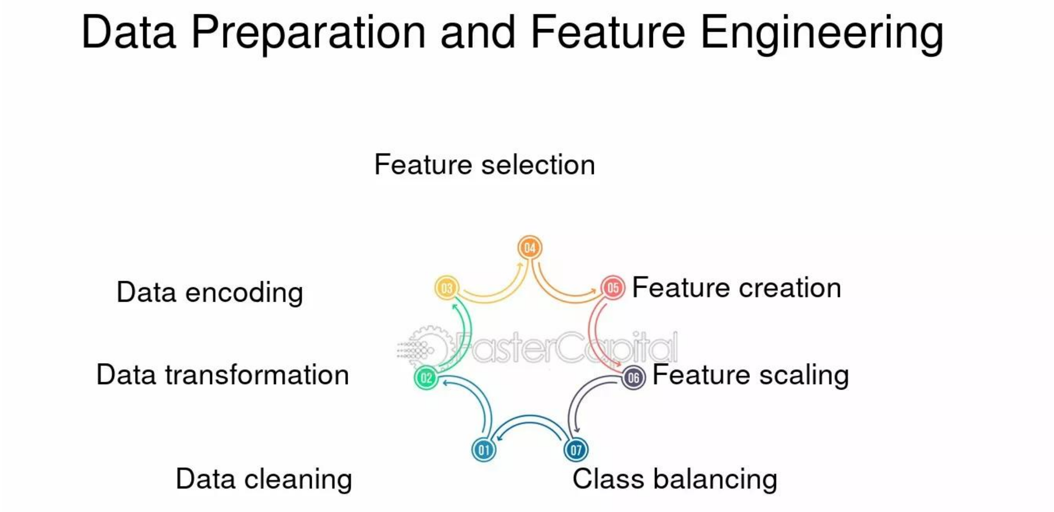
Methodology - Feature Engineering

DATA TRANSFORMATION:

- **Sentiment Aggregation:** Calculate daily or weekly sentiment averages.
- **Numerical Stock Features:**
  - Moving averages (10-day, 50-day), price volatility, and lagged returns.

FEATURE COMBINATION:

- Combine sentiment features with historical stock data to improve prediction accuracy.



Model Development

Objective: Predict stock market movements (Rise/Fall) based on the sentiment of news headlines.

Data Preprocessing:
- Text Normalization: Convert sentences to lowercase.
- Stopword Removal: Eliminate common stopwords using NLTK.
- Sentiment Mapping: Map sentiments to binary outcomes (1 = Rise, 0 = Fall).

Modeling Techniques:
- Logistic Regression: Simple linear model for binary classification.
- Random Forest Classifier: Ensemble method for improved accuracy and robustness.

Text Vectorization:
- TF-IDF: Converts text data into numerical features, limited to 5,000 features for efficiency.

Model Evaluation:
- Classification Report: Precision, recall, F1-score, and accuracy.
- Confusion Matrix: Visualizes true vs. predicted labels.

Performance Comparison:
- Test Splits: Evaluation across different train-test splits (20%, 25%, 30%).

Training and Evaluation - Optimizing the Model

TRAINING STRATEGY:

Data Split:
- 80% training, 20% testing (default)
- Different splits (20%, 25%, 30%) used for model comparison.

Optimizer:
- Logistic Regression: Default solver (e.g., LBFGS).
- Random Forest: No explicit optimizer; uses decision trees.

Metrics Used:
- **Classification Report**: Precision, Recall, F1-Score
- **Confusion Matrix**: True/False Positives/Negatives
- **Accuracy**: Proportion of correct predictions.

Classification Report (Logistic Regression):

	precision	recall	f1-score	support
0	0.79	0.97	0.87	797
1	0.88	0.45	0.60	372
accuracy			0.81	1169
macro avg	0.83	0.71	0.74	1169
weighted avg	0.82	0.81	0.78	1169

Random Forest Classification Report:

	precision	recall	f1-score	support
0	0.83	0.92	0.87	797
1	0.78	0.59	0.67	372
accuracy			0.82	1169
macro avg	0.80	0.75	0.77	1169
weighted avg	0.81	0.82	0.81	1169

Experimental Setup - Datasets and Preprocessing

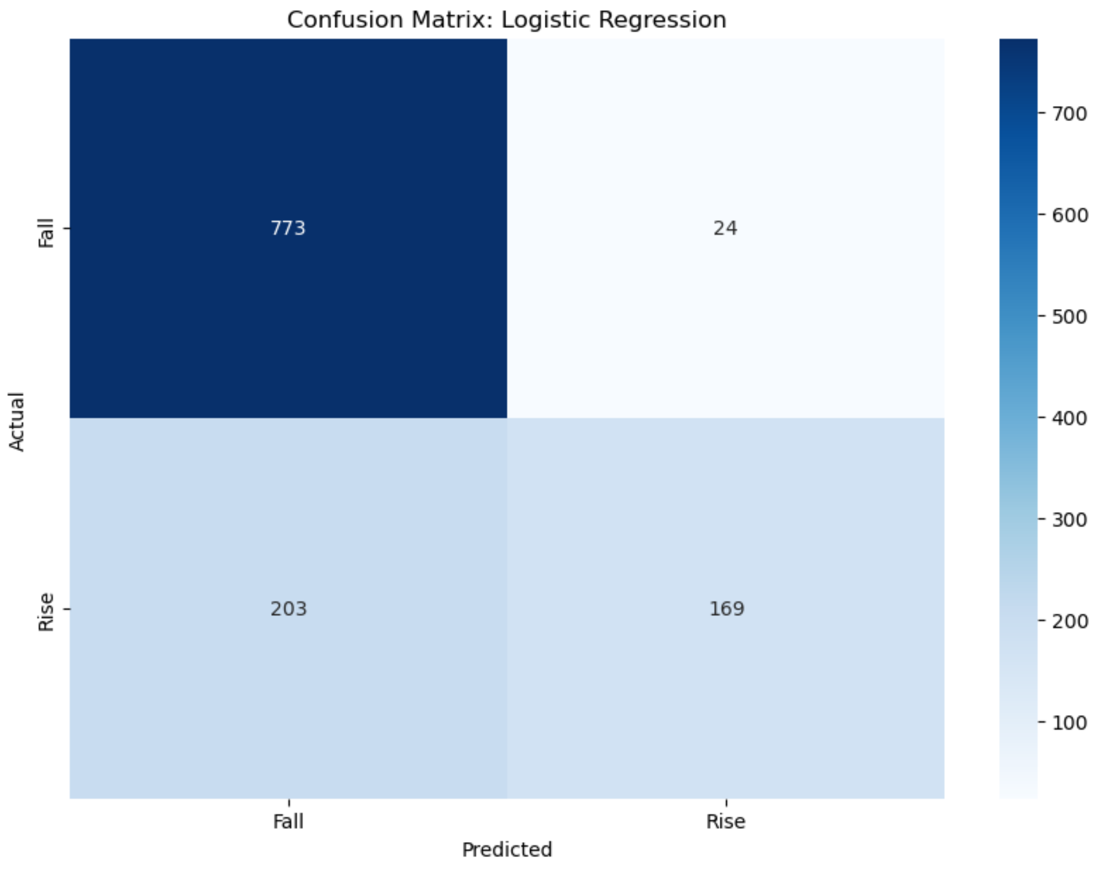
DATASETS USED:

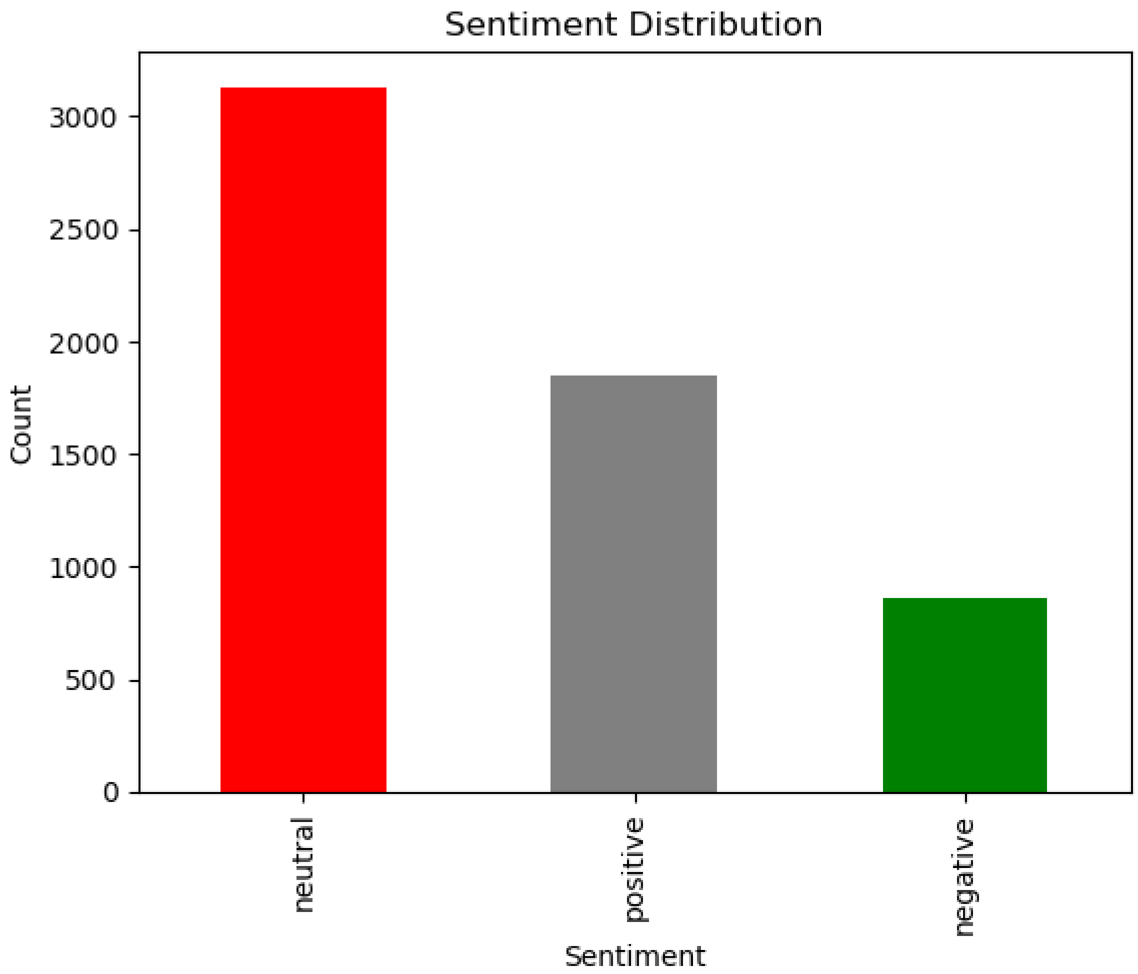
- **Financial Sentiment Analysis Dataset** (Kaggle) for headlines.
- **Yahoo Finance Data** for stock market information.

PREPROCESSING STEPS:

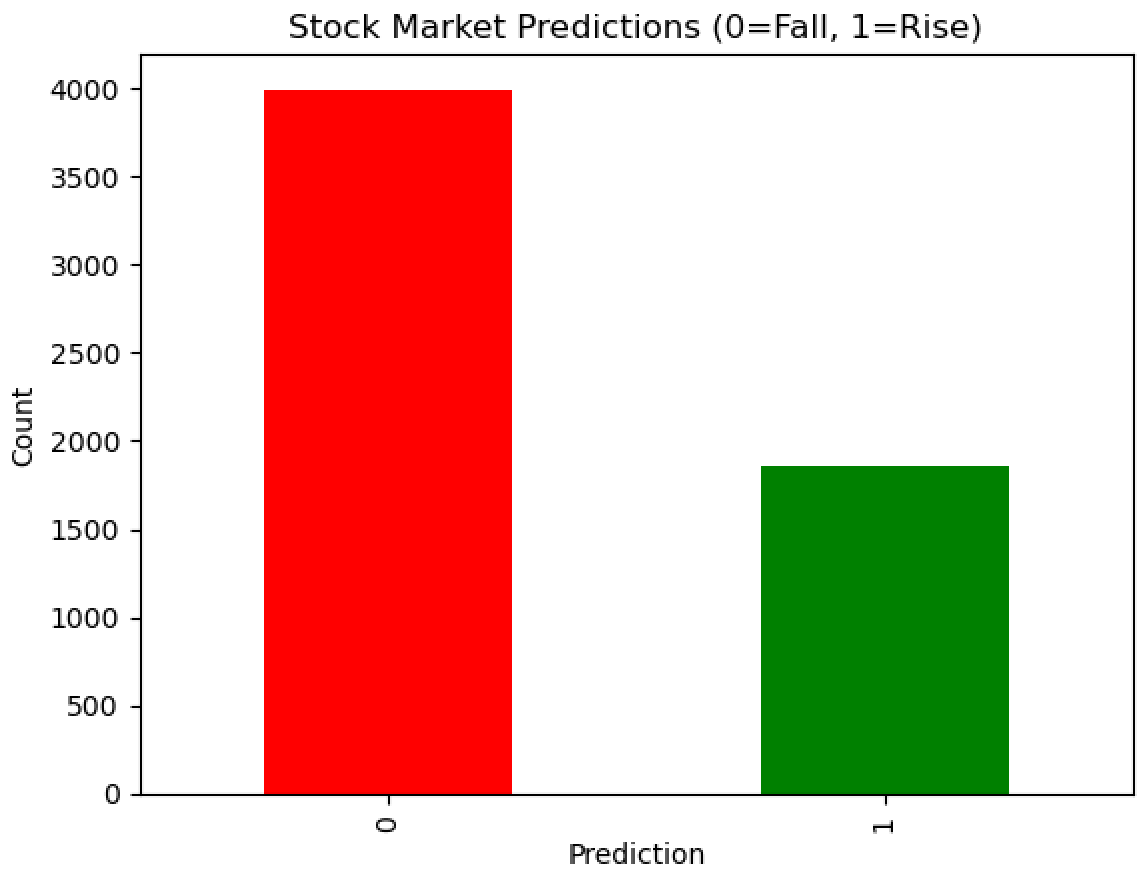
- Sentiment analysis applied to headlines.
- Stock data cleaned and merged with sentiment data.

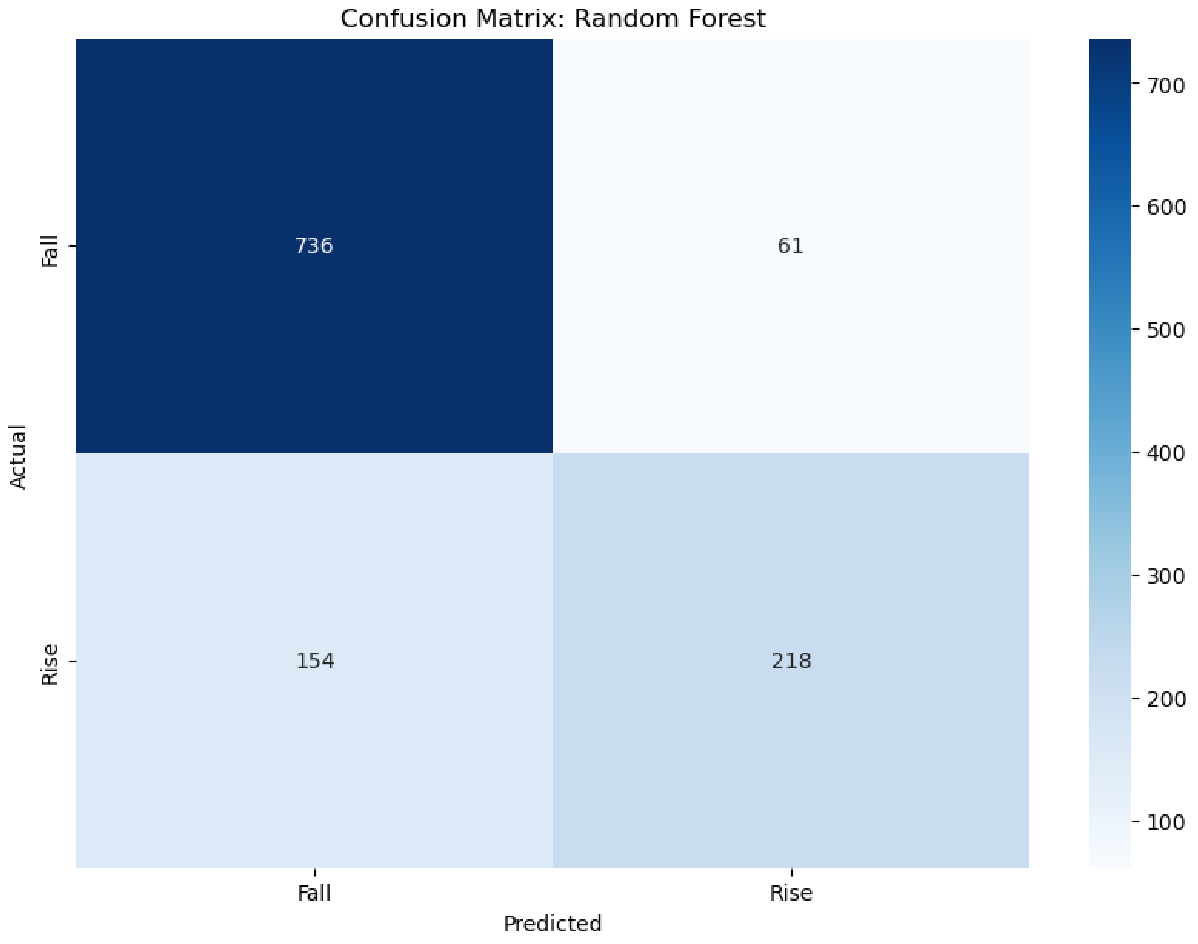
Results - Model Evaluation Metrics





Results - Model Evaluation Metrics





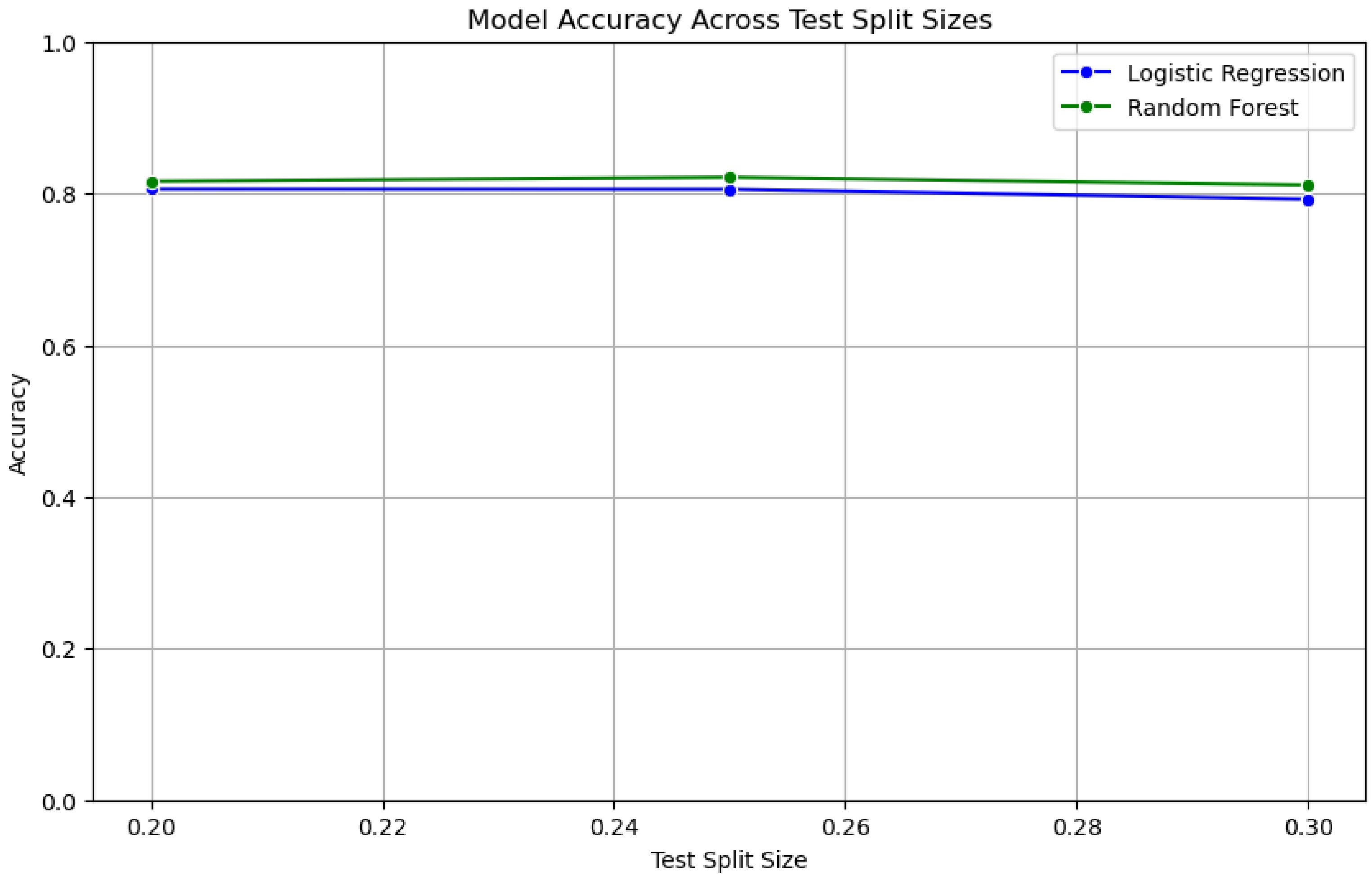
Results - Model Performance

COMPARISON OF MODELS:

- **Logistic Regression:** 81% accuracy.
- **Random Forest:** 82% accuracy.

KEY Insights:

- Positive sentiment correlates with stock price rises.



Challenges in Sentiment Interpretation

CHALLENGES FACED:

- Sentiment Ambiguity: Headlines with mixed sentiment.
- Timing Lag: Stock prices don’t always react immediately to news.
- Market Noise: Not all news affects stock prices equally.

HOW WE OVERCAME THESE CHALLENGES:

- By combining sentiment and historical stock data to create a more comprehensive predictive model.

Future Work and Conclusion

FUTURE DIRECTIONS:

1. **Expanding Data Sources:** Integrating social media sentiment from Twitter, Reddit, etc.
2. **Leveraging Advanced Models:** Exploring **BERT** and **GPT-based models** for deeper sentiment analysis.
3. **Real-Time Predictions:** Building a live pipeline for stock market predictions.

CONCLUSION:

- This research demonstrates that sentiment analysis can significantly improve stock market predictions, and we could further work on this by adding **LSTM model** will be well-suited to capture both sentiment and time-series data.

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THANK YOU